Full recursive comparison

Dataset	Model	$k = 1$	$k = 4$	$k = 16$	$k = 64$	$k = 256$
Lorenz	Ridge	0.941	0.658	0.199	0.004	0.000
	MLP	0.913 ± 0.008	0.840 ± 0.006	-0.149 ± 0.052	-0.836 ± 0.019	-0.864 ± 0.023
	Residual	0.901 ± 0.002	0.822 ± 0.005	-0.192 ± 0.013	-1.068 ± 0.069	-115.732 ± 24.623
	ODE	0.859 ± 0.009	0.794 ± 0.005	-0.372 ± 0.018	-1.241 ± 0.175	-103.849 ± 48.168
Rössler	Ridge	0.999	0.997	0.972	0.751	0.655
	MLP	0.997 ± 0.000	0.996 ± 0.000	0.994 ± 0.001	0.727 ± 0.032	0.262 ± 0.042
	Residual	0.996 ± 0.000	0.996 ± 0.000	0.987 ± 0.001	0.661 ± 0.032	-37.698 ± 8.070
	ODE	0.990 ± 0.000	0.992 ± 0.000	0.953 ± 0.016	-0.255 ± 0.615	-23.815 ± 4.710
Electricity	Ridge	0.509	0.418	0.309	0.215	0.055
	MLP	0.441 ± 0.006	0.316 ± 0.002	0.150 ± 0.007	0.064 ± 0.013	-0.014 ± 0.029
	Residual	0.373 ± 0.005	0.299 ± 0.003	0.047 ± 0.041	-0.499 ± 0.112	-22.349 ± 16.092
	ODE	0.377 ± 0.004	0.295 ± 0.005	-0.351 ± 0.129	-1.933 ± 0.201	-63.485 ± 40.650

Table 4: Same reporting starts for every model and horizon. Neural values show mean $\pm$ sample seed standard deviation. The training-target mean scores zero.

Physical timescales

$$\tau_L = \frac{1}{\lambda_{\max}}, \qquad \Delta t_{\text{analysis}} = 16\,\Delta t_{\text{raw}}.$$

Dataset	Reference clock	Horizon $k = 256$
Lorenz	$\tau_L \approx 5.8$ analysis steps	$44\,\tau_L$
Rössler	$\tau_L \approx 19.0$ analysis steps	$13.5\,\tau_L$
Electricity	Daily period: 6 analysis steps	42.7 days

Input-system Lyapunov estimates come from variational integration at the recorded parameters. They do not establish chaos in the hidden representation.

Scaling and generalization limits

Dataset	Model	Raw	Std.	White PC
Lorenz	Ridge	0.658	0.661	0.708
	MLP	0.840	0.774	0.750
Rössler	Ridge	0.997	0.994	0.996
	MLP	0.996	0.991	0.997
Electricity	Ridge	0.418	0.401	0.257
	MLP	0.316	0.287	0.100

Table 5: Four-step R^2. Std.: training coordinate scaling. White PC: whitened training PCA95. Neural entries average three seeds.

- Only four validation entities per primary dataset. Overlapping windows are not independent new series.
- About 94% of Rössler validation windows have a near-recurrent training window.
- Fixed synthetic parameters. Generalization to new parameters remains open.

Memory, observations and transfer controls

$$u_t^{\mathrm{hist}} = [h_t; h_{t-1}], \qquad u_t^{\mathrm{obs}} = [h_t; x_{q_t-15:q_t}].$$

Comparison	Forecast mode	Validation R^2
Lorenz Ridge: state / history	Direct $k = 4$	0.685 / 0.728
Lorenz MLP: state / history	Direct $k = 4$	0.806 / 0.808
Lorenz Ridge: state / observations	Direct $k = 4$	0.685 / 0.690
Frozen Lorenz Ridge on electricity	Direct $k = 1$	-4.072

These comparisons use shared eligible starts within each row. Direct k-step fits differ from recursive forecasts.

Selective gains from history do not prove a Markov state. Poor frozen transfer limits the claim of one universal fitted operator.

Early, middle and final layer controls

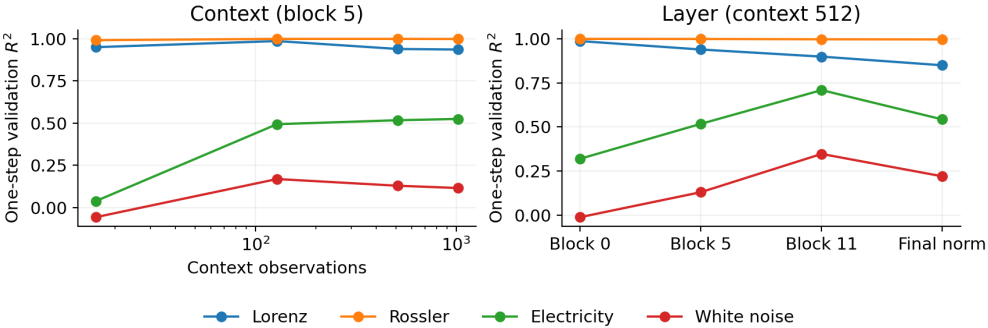


Figure 4: One-step Ridge: block 0 is first, block 5 middle (6th of 12), block 11 last. Final norm follows the last block. Each condition predicts its own hidden target.

Changing the encoder changes the target. Random-weight predictability does not measure raw-series forecasting accuracy.

Depth-control specificity

The depth predictor learns $F_d(h_{5,t}) \approx h_{6,t}$. The declared comparison scores both predictors against $h_{5,t+1}$.

$$\Delta R^2 = R^2(F_{\text{time}}) - R^2(F_d), \qquad \Delta \cos = \cos(F_{\text{time}}) - \cos(F_d).$$

- H2 requires gains of at least 0.05 in both metrics and positive paired bootstrap lower bounds.
- On white noise, even the constant training-target mean beats the depth map by 1.067 in R^2 and 0.183 in cosine.

The threshold can pass for a trivial temporal predictor. It does not establish meaningful temporal dynamics or show that depth prediction itself is difficult.

Selected references and code

- Ansari et al. Chronos-2: From Univariate to Universal Forecasting, 2025.
- Wiliński et al. Exploring Representations and Interventions in Time Series Foundation Models, 2025.
- Yang et al. From Observations to States: Latent Time Series Forecasting, 2026.
- Zhou et al. EIDOS: Latent-Space Predictive Learning for Time Series Foundation Models, 2026.
- Chen et al. Neural Ordinary Differential Equations, 2018.
- Gilpin. Chaos as an Interpretable Benchmark for Forecasting and Data-driven Modelling, 2021.

`https://github.com/nikita-aleksandrov-wr/hidden-state-evolution-analysis`